Class 7 Science Notes: Plants and Water

Photosynthesis

Photosynthesis is the process by which green plants make their own food. Plants use sunlight, water and carbon dioxide to make glucose. The green pigment chlorophyll, found in the leaves, absorbs light energy. Water is taken up from the soil by the roots and carried to the leaves. Carbon dioxide enters the leaf through tiny pores called stomata.

During photosynthesis, oxygen is released into the air as a by-product. The glucose made by the plant is used for energy and growth, and extra glucose is stored as starch. Photosynthesis takes place mainly in the leaves, inside small structures called chloroplasts.

Factors Affecting Photosynthesis

The rate of photosynthesis depends on the amount of light, the amount of carbon dioxide and the temperature. Plants kept in the dark cannot carry out photosynthesis. Very high temperatures can slow the process down because enzymes stop working well.

The Water Cycle

The water cycle is the continuous movement of water between the earth and the atmosphere. Heat from the sun causes water in oceans, rivers and lakes to evaporate and become water vapour. As the water vapour rises, it cools and condenses to form tiny droplets that make clouds. This process is called condensation.

When the droplets in clouds grow heavy, they fall back to the earth as rain, snow or hail. This is called precipitation. The water then collects in oceans, rivers, lakes and underground, and the cycle begins again. Plants also release water vapour from their leaves through a process called transpiration.

Conserving Water

Only a small part of the water on earth is fresh water that we can use. Rainwater harvesting collects and stores rainwater for later use. Fixing leaking taps and reusing water for plants are simple ways to save water at home.